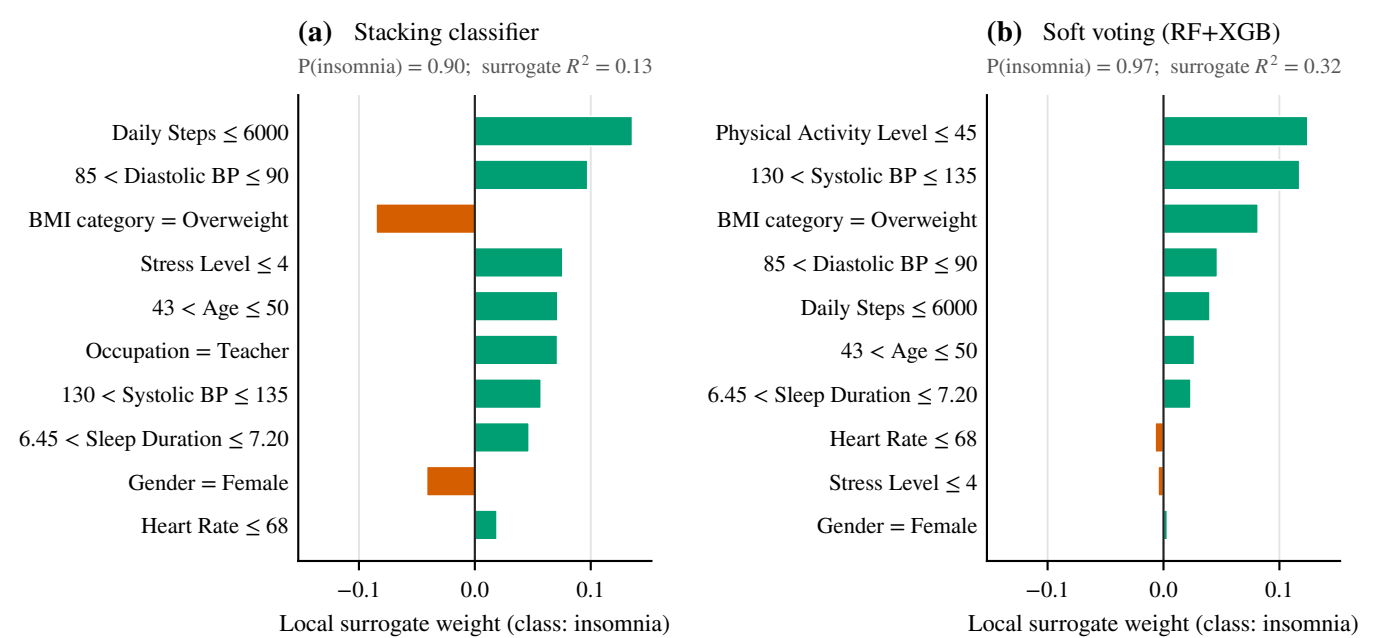
(a) Stacking classifier
P(insomnia) = 0.90; surrogate R^2 = 0.13
Daily Steps ≤ 6000
85 < Diastolic BP ≤ 90
BMI category = Overweight
Stress Level ≤ 4
43 < Age ≤ 50
Occupation = Teacher
130 < Systolic BP ≤ 135
6.45 < Sleep Duration ≤ 7.20
Gender = Female
Heart Rate ≤ 68
−0.1
0.0
0.1
Local surrogate weight (class: insomnia)
(b) Soft voting (RF+XGB)
P(insomnia) = 0.97; surrogate R^2 = 0.32
Physical Activity Level ≤ 45
130 < Systolic BP ≤ 135
BMI category = Overweight
85 < Diastolic BP ≤ 90
Daily Steps ≤ 6000
43 < Age ≤ 50
6.45 < Sleep Duration ≤ 7.20
Heart Rate ≤ 68
Stress Level ≤ 4
Gender = Female
−0.1
0.0
0.1
Local surrogate weight (class: insomnia)
raises the insomnia score
lowers the insomnia score